

CYBR 343 W13-Lab

Instructions:

- Do not delete, edit, or change any text or questions in this document.
- Type or write your answers directly below each question.
- Show all work and clearly explain your reasoning.
- Answer in complete sentences with sufficient detail. Each response must include a detailed explanation of approximately 4 to 8 lines.
- For every question, include a screenshot or snip of the artifact showing the evidence you used to support your answer.
- Place each image directly under the corresponding response.
- All images must be correctly oriented and easy to read.
- Submit your work through Blackboard only. Email submission will not be accepted.
- I will answer emails regarding this work until Friday 4:00 PM. I will not be available over the weekends. Please plan carefully.
- No credit will be given for answers that do not include both the required artifact evidence and a detailed explanation.

Objectives:

- Identify malware types discussed in slides and lecture videos.
- Interpret Volatility outputs (pslist, psscan, netscan, pstree)
- Recognize backdoors, RATs, keyloggers, and credential theft
- Apply lecture concepts to real-world forensic evidence

Provided Volatility Artifacts

Artifact 1: pslist

Offset(P)	Name	PID	PPID
0x123456	System	4	0
0x223456	smss.exe	248	4
0x323456	csrss.exe	396	248
0x423456	wininit.exe	472	248
0x523456	services.exe	560	472
0x623456	lsass.exe	620	472
0x723456	svchost.exe	1500	560
0x823456	explorer.exe	2312	1500
0x923456	chrome.exe	4020	2312

Artifact 2: psscan

Offset(P)	Name	PID	PPID
0x123456	System	4	0
0x223456	smss.exe	248	4
0x323456	csrss.exe	396	248
0x423456	wininit.exe	472	248
0x523456	services.exe	560	472
0x623456	lsass.exe	620	472

0x723456	svchost.exe	1500	560
0x823456	explorer.exe	2312	1500
0x923456	chrome.exe	4020	2312
0xA23456	svchost.exe	6124	560
0xB23456	keylog.exe	7336	2312
0xC23456	updater.exe	8120	560
0xD23456	mimikatz.exe	8452	560

Artifact 3: netscan

Proto	Local Address	Local Port	Foreign Address	Foreign Port	State	PID	Executable Path
TCP	192.168.1.10	49152	185.234.219.12	4444	ESTABLISHED	6124	C:\Users\User\AppData\Roaming\svchost.exe
TCP	192.168.1.10	49153	93.184.216.34	80	ESTABLISHED	4020	C:\Program Files\Google\Chrome\chrome.exe
TCP	192.168.1.10	49154	45.77.12.90	8080	ESTABLISHED	8120	C:\Users\User\AppData\Roaming\updater.exe
TCP	192.168.1.10	49155	185.234.219.12	9001	ESTABLISHED	7336	C:\Users\User\AppData\Roaming\keylog.exe
TCP	192.168.1.10	49156	185.234.219.12	443	ESTABLISHED	8452	C:\Users\User\AppData\Roaming\mimikatz.exe
TCP	0.0.0.0	135	0.0.0.0	0	LISTENING	560	C:\Windows\System32\services.exe

Artifact 4: pstree

System (4)

```

├── smss.exe (248)
│   └── wininit.exe (472)
│       ├── services.exe (560)
│       │   ├── svchost.exe (1500)
│       │   ├── svchost.exe (6124)
│       │   ├── updater.exe (8120)
│       │   └── mimikatz.exe (8452)
│       └── lsass.exe (620)
├── explorer.exe (2312)
│   ├── chrome.exe (4020)
│   └── keylog.exe (7336)

```

Artifact 5: Suspicious Command Execution

PID	Name	Command Line
7336	keylog.exe	C:\Users\User\AppData\Roaming\keylog.exe
8452	mimikatz.exe	mimikatz.exe privilege::debug sekurlsa::logonpasswords
8120	updater.exe	C:\Users\User\AppData\Roaming\updater.exe

Artifact 6: User Sessions

Session ID	Username	Logon Time	Source IP
1	Administrator	2023-10-12 14:10:05	LOCAL
2	User	2023-10-12 14:12:33	LOCAL
3	Admin2	2023-10-12 14:15:10	185.234.219.12
4	Guest	2023-10-12 14:16:02	45.77.12.90

Questions

Part 1: Process Analysis (20 points)

- Identify two normal system processes from **pslist** and explain why they are legitimate.
 - lsass.exe 620 472 - handles authentication, token creation, and password policy enforcement
 - services.exe 560 472 – manages system services and is always the child of wininit.exe.
 - Both processes have the correct names, expected PID, and proper parental processes

- Compare **pslist** and **psscan**.

Which processes appear only in psscan, and why is this suspicious?

- Processes that only appear in psscan and not pslist are suspicious because they were likely hidden or unlinked from the active processes list. This is typical of malware as it helps evade detection at the surface level with pslist but psscan utilizes remnants within the memory
- These processes include:
 - svchost.exe 6124
 - keylog.exe 7336
 - updater.exe 8120
 - mimikatz.exe 8452

Part 2: Malware Identification (20 points)

- Identify evidence of a keylogger in the artifacts.

Explain how it matches the behavior discussed in lecture.

- “keylog.exe 7336 2312” within C:\Users\User\AppData\Roaming\keylog.exe
- Runs from AppData\Roaming, a common malware location
- Appears only when using psscan, not pslist
- Has an active network connection using a foreign port

2. Identify a process that could act as a downloader or launcher.

Explain your reasoning using the lecture definitions.

- A process that could act as a downloader/launcher is `updater.exe`
- This process could be malicious because Malware is often disguised as updateers and communicate via ports such as 8080. It is also abnormal for an updater's parent to be `services.exe`

Part 3: Network and Backdoor Analysis (20 points)

1. Identify a suspicious connection from **netscan**.

Explain why it is suspicious.

- A suspicious connection is: PID 6124; Local Port 49152; Foreign Address 185.234.219.12; Port 4444; `svchost.exe`
- This is suspicious for the following reasons:
 1. Port 4444 is commonly used by Metasploit payloads and RATs
 2. The executable path is incorrect for an `svchost` process, it should be within `System32`, not `Roaming AppData`
 3. PID only appears using `psscan`

2. Which process is most likely acting as a **backdoor or RAT**?

Use evidence from ports, IP addresses, and process behavior.

- The most likely backdoor/RAT is `svchost.exe` PID 6124
 1. Hidden process (Revealed only using `psscan`)
 2. Fake path in `AppData`
 3. Connection to 185.234.219.12:4444
 4. Child of `services.exe`, which is abnormal
- This behavior is consistent of a RAT with persistent remote access, hidden communication with a command & control server, and hidden execution only revealed by `psscan`

Part 4: Malware Behavior and Structure (20 points)

1. Using **pstree**, explain how the malware is organized in the system.

Which process appears to be controlling or coordinating activity?

- `services.exe` (560)
 1. `svchost.exe` (6124)
 2. `updater.exe` (8120)
 3. `mimikatz.exe` (8452)
- The malware is organized under `services.exe` which shows an attacker either injected or launched the malicious process under a legitimate parent to hide. `keylog.exe` is located under `explorer.exe`, which is done to mimic user-level activities.
- `services.exe` appears to be the central controller for the malware as it spawns the `svchost` RAT, the downloader `updater.exe`, and the credential theft tool `mimikatz.exe`

2. Based on the artifacts, is this more consistent with a **RAT or botnet behavior**?

Justify your answer using lecture concepts.

- I believe this malware is more consistent with a RAT for the following reasons
 1. Direct connection to an unknown IP (Likely the attackers)
 2. Credential theft and keylogging
 3. Hidden processes under valid system parents
 4. Command execution via Mimikatz
- Unlike a botnet, this is a direct attack and the individual has been targeted specifically

Part 5: Credential Theft (10 points)

9. What type of attack is indicated by the presence of **mimikatz.exe** and its command line?

In your answer, explain:

- What the command is doing
- What type of data is being targeted
- How an attacker could use this data
 - Mimikatz.exe privilege::debug sekurlsa::logonpasswords
 - This is a credential dump attack which works by elevating privileges and extracting credentials from LSASS memory.
 - The attacker is targeting the passwords on the system, both plaintext and hashes.
 - The attacker could utilize this data to move laterally through a network, escalate privileges higher, or remote login sessions.

Part 6: Correlation and Evidence (10 points)

10. Using **sessions and network activity**, identify which user accounts are likely compromised.

Explain how the login activity and network connections support your conclusion.

- Admin2 – Source IP 185.234.219.12
- Guest – Source IP 45.77.12.90
- These IPs match the attacker controlled IPs seen in netscan, meaning that the attacker has successfully logged in using the stolen credentials via keylog.exe or Mimikatz

Part 7: Overall Analysis (10 points)

11. Describe the full attack scenario using evidence from at least three artifacts:

- Initial infection
- Hidden processes
- Remote access
- Credential theft

The attack likely began with a malicious downloader *updater.exe* that was delivered through a user action or exploit such as phishing or using a macro script in a program like word. Hidden processes discovered in psscan indicate a need for stealth for certain processes. An unauthorized remote connection was established to 185.234.129.12:4444 via a fake svchost.exe process. Keylog.exe captured keystrokes and sent them to 185.234.219.12:9001. Mimikatz.exe obtained

and dumped all available credentials from LSASS memory, which were then used to login remotely, compromising two accounts - Admin2 and Guest.